Given a system of eq^{ns} $Ax = b$

where $A \in \mathbb{R}^{k \times k}$ invertible and $b \in \mathbb{R}^k$, solve for x.

$\tilde{L} A x = \tilde{L} b$

$Ux = \tilde{L} b$

$U = \begin{bmatrix} * & * & \cdots & * \\ & * & \ddots & \vdots \\ & & \ddots & * \\ 0 & & & * \end{bmatrix} \in \mathbb{R}^{k \times k}$

$\tilde{L} A = U \Rightarrow A = LU$ where $L = \tilde{L}^{-1}$

L is a lower triangular matrix with all 1's on the diagonal & U is a upper triangular matrix.

The LU decomposition is unique.

$A = L_1 U_1 = L_2 U_2 \Rightarrow L_1 L_2^{-1} = U_2 U_1^{-1} \overset{??}{\Rightarrow} L_1 = L_2$ & $U_1 = U_2$ ■

If $\hat{\alpha}$ is the solution to the LS problem:

$$\min_{\alpha \in \mathbb{R}^k} \| \Phi \alpha - y \|_2^2$$

if and only if $\hat{\alpha}$ satisfies

$$\Phi^T \Phi \hat{\alpha} = \Phi^T y \quad : \text{normal eq}^{ns}$$

i.e. in order to find the solution to the LS problem, we have to solve:

$$\Phi^T \Phi \alpha = \Phi^T y$$

$\Phi \in \mathbb{R}^{N \times k}$, $y \in \mathbb{R}^N$, $\alpha \in \mathbb{R}^k$

$\Phi \in \mathbb{R}^{N \times k}$ has all the columns lin. indep. ($\operatorname{rank} \Phi = k$)

$\Leftrightarrow$ $\Phi^T \Phi$ is invertible.

$\Phi^T \Phi$ is invertible.

Suppose on contrary, $\Phi^T \Phi$ is NOT invertible, there exists a non-zero $x \in \mathbb{R}^k$ s.t.

$$\Phi^T \Phi x = 0$$

$$\langle x, \Phi^T \Phi x \rangle = x^T \Phi^T \Phi x = 0$$

$$\Rightarrow (\Phi x)^T \Phi x = 0$$

$$\Rightarrow \|\Phi x\|_2^2 = 0$$

$$\Rightarrow \Phi x = 0$$

$x = x_1 c_1 + x_2 c_2 + \cdots + x_k c_k$

$\Phi x = \Phi(x_1 c_1 + \cdots + x_k c_k) = x_1 (\text{first column of } \Phi) + \cdots + x_k (k^{\text{th}} \text{ column of } \Phi)$

$= 0$

$\{f_1, f_2, \dots, f_k\}$ is linearly independent set.

$$\Rightarrow \Phi = \begin{bmatrix} f_1(x_1) & f_2(x_1) & \cdots & f_k(x_1) \\ f_1(x_2) & f_2(x_2) & & f_k(x_2) \\ \vdots & \vdots & & \vdots \\ f_1(x_N) & f_2(x_N) & \cdots & f_k(x_N) \end{bmatrix} \in \mathbb{R}^{N \times k}$$

columns of Φ are linearly independent.

$f_i : \mathbb{R}^n \to \mathbb{R}$, $x_1, \dots, x_N \in \mathbb{R}^n$

$y_1, \dots, y_N \in \mathbb{R}$

$\beta_1 f_1 + \beta_2 f_2 + \dots + \beta_k f_k = 0 \Rightarrow \beta_1 = \beta_2 = \dots = \beta_k = 0$

$(\beta_1 f_1 + \dots + \beta_k f_k)(x_i) = 0$ $\forall\, x_i \in \mathbb{R}$

$\Rightarrow \beta_1 = \dots = \beta_k = 0$

?? $\rightarrow$ $\begin{bmatrix} f_1(x_i) & f_2(x_i) & \cdots & f_k(x_i) \end{bmatrix} \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} = 0$

for $i = 1, 2, \dots, N$

$$\left\{ f_1 = 1,\ f_2 = x,\ \cdots,\ f_k = x^{k-1} \ \text{on} \right.$$
$$\left. N >> k \right\}$$

$$A = \Phi^T \Phi \quad \in \mathbb{R}^{k \times k}$$

$$b = \Phi^T y \quad \in \mathbb{R}^k$$

$A = \Phi^T \Phi$ is invertible.

A : symmetric.

$$\Phi^T \Phi v = \lambda v \qquad \lambda \in \mathbb{R},\ \overset{0}{\neq} v \in \mathbb{R}^k$$

$$\|\Phi v\|_2^2 = v^T \Phi^T \Phi v = \lambda v^T v \quad \Rightarrow \lambda > 0$$

$Ax = b$

$\Phi^T \Phi$ invertible.

$\Phi^T \Phi v = \lambda v$

if $\lambda = 0$, then

$\Phi^T \Phi v = 0$

contradict to

$\Phi^T \Phi$ invertible.

$\Phi^T\Phi$ is positive definite. { – square, – symmetric, – all e.values > 0. }

$$A = \Phi^T\Phi = LU$$

$$= LD\tilde{U}$$

D = diagonal matrix

$$= \begin{bmatrix} u_{11} & & 0 \\ & u_{22} & \\ & \ddots & \\ 0 & & u_{nn} \end{bmatrix},$$

$$\tilde{U} = \begin{bmatrix} 1 & & * \\ & 1 & \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

$LD\tilde{U}$

$=$

$(LD\tilde{U})^T = \tilde{U}^T D L^T \Rightarrow L = \tilde{U}^T$

A

$= \tilde{U}^T D \tilde{U}$

Cholesky decomposition of A

$A = \tilde{U}^T D^{1/2} D^{1/2} \tilde{U}$

$Ax = b$

→ LU / Cholesky decomposition.

$$\tilde{L}Ax = \tilde{L}b$$

$$\underbrace{LA}\,x = \tilde{L}b$$

$$Ux = \tilde{L}b$$

→ QR decomposition.

$Q \in \mathbb{R}^{k \times k}$ orthogonal.

$$Q^T Q = Q Q^T = I_k$$

$$Q = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \cdots & q_k \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{k \times k}$$

$$Q^T Q = I$$

$$e_i^T Q^T Q e_i = \|Q e_i\|_2^2$$

$$= \|q_i\|_2^2$$

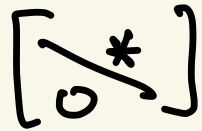
$Ax = b$

$Q^T A x = Q^T b$

$Rx = Q^T b$

↑

upper triangular



=

R

$Q^T A = R \Rightarrow \underline{A = QR}$

Q: orthogonal.

R: upper triangular.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Find $Q^T \in \mathbb{R}^{2\times 2}$ s.t.

$$Q^T A = \begin{bmatrix} * & * \\ 0 & * \end{bmatrix} = R$$

$$\left\| \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} \right\|_2^2 = a_{11}^2 + a_{21}^2$$

$\pm\sqrt{a_{11}^2 + a_{21}^2}$

$$Q^T \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} * \\ 0 \end{bmatrix}, \quad Q^T \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} = \begin{bmatrix} * \\ * \end{bmatrix}$$

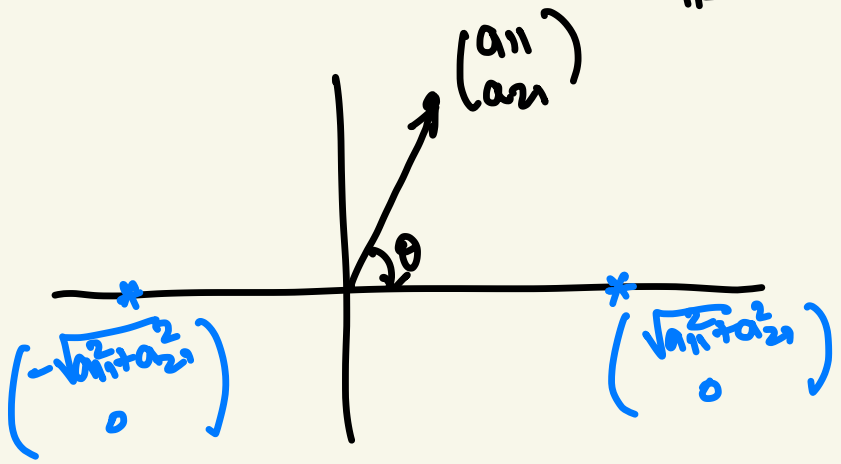
Q is orthogonal matrix.

Then $\|x\|_2^2 = \|Qx\|_2^2 = (Qx)^T Qx = x^T Q^T Q x = x^T x$

$\|$

$x^T x$

$\mathbb{R}^2$



Is rotation of vector in (anti) clockwise direction a lin. transformation?

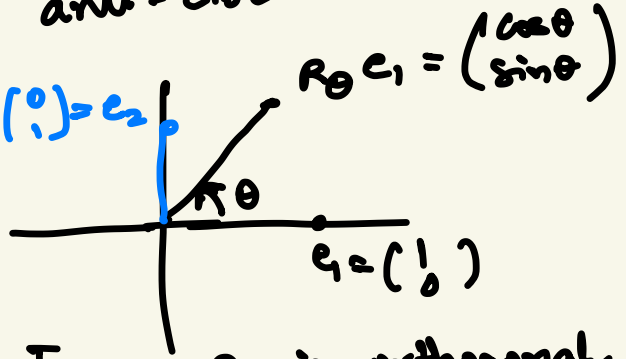
Lin. transformation: $T : \mathbb{R}^2 \to \mathbb{R}^2$

R_θ : rotation in anti-clockwise direction.

$$R_\theta e_1 = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

$$R_\theta e_2 = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$$

$$\Rightarrow R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$



$R_\theta^T R_\theta = I_2 \Rightarrow R_\theta$ is orthogonal.